KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: HOSPITAL PLANT & BUILDING SERVICES THEORY

DATE: 14th June 2022	**TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

HOSPITAL PLANT

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. Which one of the following is not true about use of copper pipe in gas installation?
   a) Copper pipes can be joined quickly
   b) Copper pipes have smooth internal bore
   c) Copper pipes are flexible
   d) Hemp sealing can be used I copper pipe gas joints

2. Which of the following appliances does not discharge soiled water?
   a) Water closet
   b) Urinal
   c) Wash hand basin
   d) None of the above

3. The gas passes through injector nozzle and draws air with it. Which type of gas burner is described as above?
   a) Non generated
   b) Aerated
   c) Gas cookers
   d) Gas water heaters

4. A two-stroke engine uses more gasoline and burns a lot of oil, so it is far more
   a) Polluting
   b) Expensive
   c) Stable
   d) Unstable

5. Which one below is the main activity that takes place in hydro-extractor
   a) Spinning, Breaking, Unlocking
   b) Washing, Heating, Unlocking
   c) Washing, Spinning, Breaking
   d) Spinning, Washing, Unblocking

6. The two pumps used for pressure creation in pneumatic system are?
   a) Piston and rotary
   b) Centrifuge and piston
   c) Rotary and centrifuge
   d) Piston and centrifuge pump

7. Which one of the following is odd one out?
   a) Sedimentation
   b) Reverse osmosis
   c) Filtration
   d) Calcification

8. In water supply mains, water hammer is eliminated by providing
   a) Air valve
   b) Pressure relief valve
   c) Hydro-pneumatic accumulator
   d) Stop valve

9. The following are components of absorption refrigeration system Except:
   a) Evaporation
   b) Condenser
   c) Reservoirs
   d) Filter drier

10. In compression refrigeration system the condition of refrigerant is saturated liquid
   a) After passing through the condenser
   b) Before passing through the condenser
   c) After passing through expansion throttle valve
   d) Before entering the expansion valve

11. Endoscopy is one of the best techniques to enter and view different areas of the body. Which mechanical science principle does the endoscope use:
   a) Total internal reflection of light
   b) Reflection and refraction of light
   c) Propagation of light
   d) Absorption and re-emission of light

12. Which one of the following is a part of a heavy diesel burner?
   a) Pressure switch
   b) Injector nozzle
   c) Pressure gauge
   d) Bourdon gauge

13. Which gas law depicts the relationship between the volume of a fixed mass of a gas and its pressure at a constant temperature?

   a) Charles Law
   b) Dalton Law
   c) Boyle's Law
   d) Gay Lussac's Law

14. When a piping system is made up primary of friction head and very little of vertical lift then the pump characteristics should be
   a) Horizontal
   b) Steep
   c) First rise and then fall
   d) Nearly horizontal

15. To ensure human comfort the air-conditioner should be set at what range of temperature?
   a) $22^0c - 26^0c$
   b) $30^0c - 43^0c$
   c) $35^0c - 38^0c$
   d) $28^0\ c - 32^0\ c$

16. Absorption refrigeration system normally uses the following refrigerant
   a) Freo-11
   b) Freon-22
   c) Carbon dioxide
   d) Ammonia

17. The ignition coil if gasoline engine is a transformer. Which type of transformer is it?
   a) Step-up transformer
   b) Step-down transformer
   c) High current transformer
   d) Low voltage transformer

18. Which of the following taps are used on wash hand basins?
   a) Stop valves
   b) Bib taps
   c) Stop cocks
   d) Pillar taps

19. Steel is an alloy of:
   a) Iron and aluminium
   b) Copper and iron
   c) Carbon and iron
   d) Aluminium and carbon

20. Which of the following is not a type of positive displacement pumps?
   a) Reciprocating pumps
   b) Rotary displacement pumps
   c) Centrifugal pump
   d) None of the above

21. The lowest temperatures during the cycle in a vapour compression refrigeration system occurs after:
   a) Compression
   b) Evaporation
   c) Condensation
   d) Expansion

22. Which of the following describes the functions of the trap in sanity appliance?
   a) Trap the condensate
   b) Trap dirty water
   c) Keep foul smell
   d) Prevent dirt from accumulating in sanity appliances

23. The accumulator in a domestic refrigeration is connected between 'which major components of the refrigerator:
   a) Compression and condenser
   b) Capillary tube and condenser
   c) Evaporator and condenser
   d) Evaporator and compressor

24. Which components from ones listed below releases excess pressure to the reservoir in hydraulic system
   a) Non-return valve
   b) Throttle valve
   c) Relief valve
   d) Direction control valve

25. Lens which converges incident rays on a single point is:
   a) Convex lens
   b) Concave lens
   c) Biconcave lens
   d) Both B and C

26. Engine pistons are usually made of aluminum alloy because it
   a) Is lighter
   b) Wears less
   c) Absorbs shocks
   d) Is stronger

27. The mechanical workshop machine, which is used in the production of gears and precision machines parts is called:-
   a) Lathe machine
   b) Milling machine
   c) Planning machine
   d) Boring machine

28. The purpose of tumbler drier is to facilitate
   a) Washing
   b) Spinning
   c) Drying
   d) Disinfection

29. Which one of the following statement is true?
   a) Diesel engine has got fuel injector
   b) Diesel engine has a lower compression ratio that petro engine
   c) The operation cot of the diesel us higher than that of a petrol engine
   d) Diesel engine is s two stroke cycle engine

30. In electric cookers heat intensity is controlled by
   a) Summer start
   b) Heat controller
   c) Control knob
   d) Regulator knob

31. Which material is used as cutting tool in a mechanical workshop
   a) Carbide
   b) Iron
   c) Copper
   d) Ceramics

32. The following are rotary pump except

a) Gear pumps
b) Vane pumps
c) Screw pumps
d) Plunyer pumps

33. Which protocol was revised and strengthened in 1992 to stop the manufacture of chlorofluorocarbon refregements
a) Montreal protocol
b) Kyoto protocol
c) Geneva protocol
d) Ottawa protocol

34. What is the disadvantage of ammonia as a refrigerant?

a) Ammonia cannot be detected in case of leakage
b) Ammonia has bad effect on ozone layer
c) Ammonia is toxic in nature
d) Ammonia has higher energy cost

35. A pump which develops a constant output is called:-
a) A high frequency pump
b) A positive displacement pump
c) A variable volume pump
d) An impeller pump

36. Which one of the following components is not found in a hydraulic system
a) Pump
b) Actuator
c) Compressor
d) Motor

37. Which one of the following medical equipment uses hydraulic and pneumatic system
a) Dental chair unit
b) Operating theatre light
c) Anaesthetic machine
d) Delivery bed

38. Which one of the following boilers is high pressure boiler?

a) Water tube boiler
b) Fire tube boiler
c) Electric boiler
d) All of the above

39. The ability of a material to deform under compressive load is called
a) Ductility
b) Malleability
c) Plasticity
d) Elasticity

40. A device used to remove condensate from steam distribution system is called

a) Vent
b) Seal
c) Steam trap
d) Steam valve

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41 As the section head of the mechanical workshop in the hospital you are required to oversee proper plumbing that will ensure adequate water supply in the hospital with minimum maintenance. What are some of the key areas to note on the plumbing fixtures and pipes so as ensure good workmanship? 5 marks

42 A laundry attendant at Kiambu level 4 hospital complains that the washing machine is not spinning. What could be the probable cause to the problem? 5 marks

43 A motor driven juice blender fails to operate when the put on power. Assuming that the blender has no power indicator, state the possible causes of failure? 5 marks

44 Name any Five uses of boilers for medical steam in hospitals 5 marks

45 Explain the functions of the following components in a vapour compression system 5 marks

i)Compressor

ii)Condenser

iii)Filter drier

iv)Evaporator coil

v)Cold thermostat

46 How is dehumidification achieved by split system in air conditioning? 5 marks

47 State some of the monthly checks done on system efficiency on cold room refrigeration system 5 marks

48 State Five types of evaporators used in vapour compression refrigeration system. 5 marks

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49 a As the biomedical technician in charge of a level three hospital, the hospital management team sorts for your technical expertise on the purchase of medical refrigerators to be used for the storage of medicines and vaccines in the hospital. What are some of the factors that you should consider for the purchase of these medical refrigerators? 14 marks

b What are some of the maintenance carried out on compressor units after every six months? 6 marks